

V19 FX PROP DESK — SYSTEM ARCHITECTURE

Hierarchical Data Flow: MT4 → FastAPI → 6 Engines → PostgreSQL + Redis → Dashboard

MT4 TERMINAL

DataCollector.mq4

- › Runs on H1 bar close — polls 6 symbols
- › Collects: OHLCV + EMA50/200, RSI14, ADX14
- › Collects: DI+(Plus), DI−(Minus), ATR14
- › HTTP POST → FastAPI /data/candle (JSON)
- › Timer-based: EventSetTimer(60s)

TradeExecutor.mq4 (v2 — all fixes applied)

- › CB: one-way daily latch (realized + floating loss)
- › Queries /analytics/drawdown — not just balance delta
- › Friday ≥20:00 UTC: CloseAll + block new entries
- › Signal poll: /signals/evaluate/{sym}?pip_value=live
- › Partial-close: per-ticket latch array (never fires twice)
- › BE = entry ± 1-pip buffer; ATR trail after +2R
- › Telegram alerts, CB / trade-open / close-all

FASTAPI BACKEND (port 8000)

/data

- › POST /candle
- › GET /candles/{sym}

/signals

- › GET /evaluate/{sym}
- › GET /circuit-breaker

/trades

- › POST /open
- › POST /close/{id}
- › POST /manage/{id}

/analytics

- › GET /performance
- › GET /drawdown
- › GET /equity-curve
- › GET /trade-history

evaluate / gate / read / write

6 BUSINESS ENGINES

P1

SIGNAL ENGINE

- › EMA50 > EMA200 trend
- › Regime: EMA gap ≥ 0.1%
- › Body ≤ 1.5 × ATR guard
- › HTF H4 bias (optional)
- › RSI 55-70 / 30-45
- › ADX>40 → RSI→80/20
- › DI+ > DI− (BUY)
- › ATR > minimum floor

P2

RISK ENGINE

- › Risk: 0.25% equity/trade
- › SL: ATR × 1.5
- › TP: ATR × 3.0 (2R:1R)
- › Live pip value from MT4
- › Spread guard ≤ 2× avg
- › Lot rounded to 0.01
- › CB: 3% / 6% / 10% DD
- › Realized + float loss

P3

PORTFOLIO

- › Max 3 open positions
- › Corr filter: 60-period
- › Max corr threshold: 0.80
- › Currency overlap removed
- › Exposure: USD-normalised
- › Limit: Equity × 2

P4

NEWS ENGINE

- › TradingEconomics API
- › Fail-safe block (default)
- › 15min cache if <2h away
- › 1hr cache otherwise
- › HIGH impact only
- › ±30min blackout window

P5

SESSION ENGINE

- › London 08–17 UTC
- › New York 13–22 UTC
- › Overlap 13–17 UTC
- › DST-aware (pytz)
- › Friday ≥20:00 → close
- › Weekend detection

P6

TRADE MANAGER

- › +1R → BE + 1pip buffer
- › Early trail option (1R)
- › +2R → close 50% once
- › Per-ticket latch array
- › +2R → ATR×1.5 trail
- › NONE if no condition met

DATA STORE

PostgreSQL (port 5432)

- › market_data — OHLCV + 7 indicators, symbol/TF indexed
- › trades — open positions: entry, SL, TP, lot, session, equity
- › trade_history — closed trades: R-multiple, net PnL, exit reason
- › quant_journal — daily equity / DD / WR / PF / Sharpe
- › news_events — HIGH-impact calendar cache
- › db-backup service: daily pg_dump → ./backups/ (30-day TTL)

Redis (port 6379)

- › news:{date} — HIGH-impact events (15min or 1hr TTL)
- › Dynamic TTL: shorten to 15min when event < 2hrs away
- › Circuit breaker state — persists across server restarts
- › Fast read path for signal evaluation hot loop

persist to PostgreSQL

ANALYTICS DASHBOARD (index.html)

Equity Curve

- › Chart.js line chart
- › Full account history

Open Positions

- › Real-time table
- › Entry/SL/TP/Session

Daily P&L

- › Bar chart per day
- › Green/red colouring

Performance KPIs

- › Win Rate / Profit Factor
- › Sharpe / Expectancy

Drawdown

- › Daily/Weekly/Monthly
- › CB threshold markers

Trade History

- › All closed records
- › R-multiple / exit reason